Data analysis for experimental laboratories

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I've written this "little red book" from memory and as a tribute to a similar one given to me as an undergraduate at the University of Durham in 2005 by someone I remember as Dr. Tom.

Of course, the errors present are my own. I hope this version will stand students in good stead, as Dr. Tom's one did for me.

Subsequent information, for which I am indebted to Prof. Tom Shanks, led me to understand that the original "little red book" was written by Prof. Ifan Hughes and Prof. Tom Hase. Their booklet became the basis of their textbook,

Ifan Hughes and Tom Hase. Measurements and their Uncertainties: A practical guide to modern error analysis. *Oxford University Press, 2010.* ISBN*: 978-0-19-956632-7*

which I thoroughly recommend to you.

Andrew Fowlie. Suzhou. 2026

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1 Physical constants

These are physical constants in the International System of Units (SI).

Constant	Symbol	Value
Speed of light in vacuum	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s}$
Elementary charge	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{m^{3}\,kg^{-1}\,s^{-2}}$
Boltzmann constant	k	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}}$
Avogadro's number	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
Electron mass	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Proton mass	m_p	$1.672 \times 10^{-27}\,\mathrm{kg}$
Electric constant	ε_0	$8.854 \times 10^{-12}\,\mathrm{F/m}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Fine-structure constant	α	$1/137$
Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$

2 Formulas & notation

2.1 SI prefixes

These are prefixes for the International System of Units (SI). They are placed before a unit. E.g., $1\,\mu m = 10^{-6}\,m$ and $1\,GJ = 10^{9}\,J$.

Prefix	Symbol	Multiplier
tera	T	10^{12}
giga	G	10^{9}
mega	M	10^{6}
kilo	k	10^{3}
hecto	h	10^{2}
deca	da	10^{1}
deci	d	10^{-1}
centi	c	10^{-2}
milli	m	10^{-3}
micro	μ	10^{-6}
nano	n	10^{-9}
pico	p	10^{-12}
femto	f	10^{-15}
atto	a	10^{-18}

2.2 Scientific notation

It's inconvenient to write very large or very small numbers in decimal notation. E.g.,

$$0.00000000000000034343$$

or

$$9834333000000$$

are long and hard to read. Instead, we write them in scientific notation in the form

$$m \times 10^n$$

where n is an integer called the exponent and m is a real number called the mantissa. In so-called standard form, m and n are chosen such that $1 \leq |m| < 10$. E.g., the two numbers given previously can be written compactly in standard form as,

$$3.4343 \times 10^{-16}$$

and

$$9.834333 \times 10^{12}$$

When the exponent is zero, it is acceptable to omit the factor 10^0, e.g., 3×10^0 may be written simply as 3.

In computer programs and calculators, scientific notation is often written using a capital `E`. E.g., `3E-5` means 3×10^{-5}.

2.3 Significant figures

When we communicate a number, we should only communicate digits that are reliably known and meaningful. E.g., it makes no sense to tell you that my height measured by a tape measure is

$$1.82200000\,\mathrm{m}$$

This is poor and misleading communication — I didn't measure my height with such precision. If I can measure to about the nearest mm, I should instead tell you,

$$1.822\,\mathrm{m}$$

since now all digits are meaningful and convey information.

This is an example of significant figures (or sig. figs.). If we write a number to k significant figures, it means that we round the number to k digits and only report them. When communicating any number, we should think carefully about the appropriate number of significant figures — that is, the number of digits that are meaningful and known reliably.

If we write numbers in standard form, the number of significant figures is the number of digits in the mantissa. Zeros can be significant. E.g.,

$$1.200 \times 10^{12}$$

contains 4 significant figures, whereas

$$1.2 \times 10^{12}$$

contains only 2 significant figures. If we do not write numbers in standard form, the number of significant figures can be ambiguous. E.g., how many significant figures are there in 120? It could be 2 or 3, depending on whether the trailing zero was significant.

Finally, when writing a result to a particular number of significant figures, any rounding should be done only on the final result. E.g., suppose we wish to write cr with $r = 0.95\,\mathrm{m}$ to 2 significant figures. The result would be

$$2 \times \pi \times 0.95\,\mathrm{m} = 5.969\ldots\,\mathrm{m} = 6.0\,\mathrm{m}\ (2\ \text{sig. fig.})$$

If we incorrectly rounded π to 2 significant figures, we would instead obtain

$$2 \times 3.1 \times 0.95\,\mathrm{m} = 5.89\,\mathrm{m} = 5.9\,\mathrm{m}\ (2\ \text{sig. fig.})$$

2.4 Dot & cross products

We regularly need to manipulate vectors in physics. When doing so, there are two important products of two vectors.

Dot product

First, the dot or scalar product of two vectors, which returns a scalar,

$$\vec{a} \cdot \vec{b} = \sum_i a_i b_i = |a||b| \cos\theta$$

The dot product is commutative, that is, $\vec{a} \cdot \vec{b} = \vec{b} \cdot \vec{a}$. If two vectors are perpendicular, their dot product is zero.

Cross product

Second, the cross or vector product between two vectors. The cross product

$$\vec{a} \times \vec{b} = \vec{c}$$

returns a vector $\vec{c}$ that is perpendicular to both $\vec{a}$ and $\vec{b}$ with magnitude,

$$|c| = |a||b| \sin\theta.$$

The direction of the vector $\vec{c}$ depends on a choice of coordinate system. In physics, we use a right-handed coordinate system. The direction of the vector $\vec{c}$ is thus determined by the right-hand rule; see fig. 1.

In component notation and in a right-handed coordinate system,

$$\vec{c} = \vec{a} \times \vec{b} = \begin{bmatrix} a_2 b_3 - a_3 b_2 \\ a_3 b_1 - a_1 b_3 \\ a_1 b_2 - a_2 b_1 \end{bmatrix}$$

The cross product is anti-commutative, that is, $\vec{a} \times \vec{b} = -\vec{b} \times \vec{a}$. If two vectors are parallel, their cross product is zero.

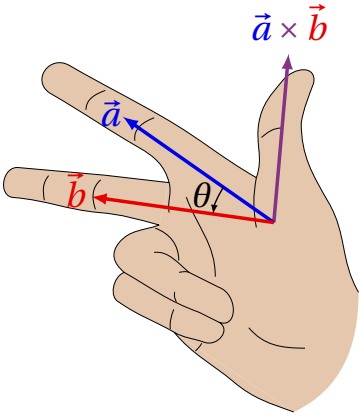


Figure 1: *Right-handed rule for the direction of the cross product between two vectors.*

3 Experiments and uncertainties

Broadly speaking, experiments in physics can be either measurements or tests. In a measurement, we want to find, e.g., the mass of the electron, the speed of light, or the period of a pendulum. In a test, we want to check whether a theory or principle works, e.g., momentum conservation or the universality of free fall.

In any case, our equipment, our understanding and control of our equipment and our control of the experimental conditions are always imperfect. There are thus always errors or uncertainties in our experiments. In physics, the words uncertainty and error are used somewhat interchangeably. The word error doesn't imply a mistake or blunder.

Without an assessment of uncertainty, measurements are almost worthless and tests are impossible. For example, suppose I reported that a car's velocity was $10\,\mathrm{ms}^{-1}$ but didn't tell you that my uncertainty was $20\,\mathrm{ms}^{-1}$. We can't in fact even be sure which direction the car is traveling. Suppose I invented a machine that created 500 J of energy, violating energy conservation, but didn't tell you that my uncertainty was 1000 J. The machine

is in fact consistent with energy conservation.

3.1 Precision and accuracy

Consider shooting a bunch of arrows at a target, aiming for the center. There are two ways we could miss. First, our arrows could be imprecise — they could be scattered around the target, but at least focused around the center. Second, our arrows could be inaccurate — they could be tightly grouped together, but missing the center of the target. These possibilities are sketched in fig. 2.

Similarly, our measurements of an unknown quantity could be imprecise or inaccurate. These discrepancies are caused by statistical and systematic errors, respectively.

3.2 Statistical and systematic errors

Statistical errors are caused by random scatter of a measurement about the true unknown value. They diminish as we collect more data — by repeating our measurements n times, we may reduce our statistical errors by a factor $1/\sqrt{n}$.

For example, suppose we time the period of a pendulum using a stop watch. We cannot start and stop the watch at exactly the moment the pendulum crosses a particular point — this is a coincidence error. Each time we try, we record a different time. There is thus a statistical component to the uncertainty. In this

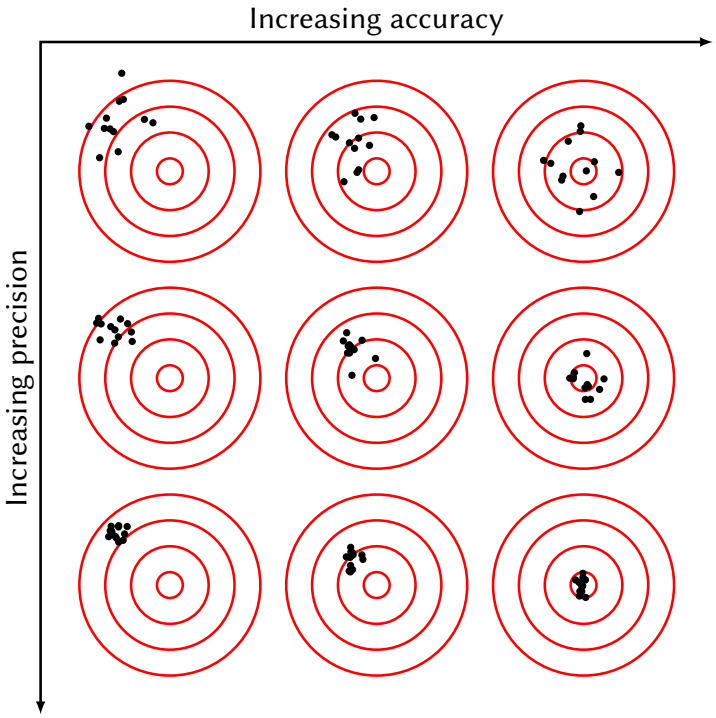


Figure 2: *Precision versus accuracy.*

case, the impact of the coincidence error may be reduced by timing several periods and choosing a fiducial marker where the pendulum is moving fast. And the remaining statistical error maybe reduced by taking repeat measurements.

Systematic errors, on the other hand, are caused by our lack of complete understanding of our experiment, including e.g., potential faults in our equipment, misuse of our equipment, and simplifications and approximations in our modeling. E.g., consider a parallax error, sketched in fig. 3. When making a measurement, our eye must be perpendicular with the scale. If it isn't, our measurements are systematically shifted by our viewing angle.

Systematic errors cause a bias or non-random deviation, e.g. a constant shift, in our measurement from the true unknown value. They cannot be averaged away by collecting more data. They should, however, be minimized in our experimental design. E.g., by being aware of and avoiding parallax errors, calibrating any equipment, taring scales, or by shielding equipment from environmental noise.

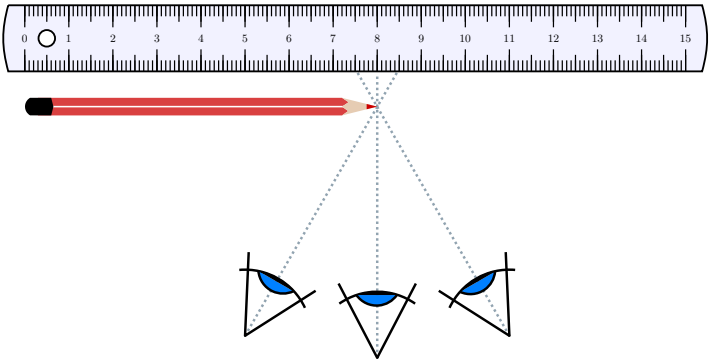


Figure 3: *Parallax error when measuring with a ruler.*

4 Quantifying uncertainties

4.1 Resolution

Experimental data are often read from an instrument with an analogue scale (e.g., a ruler) or a digital scale (e.g., a stop watch). These measurements are limited by resolution, that is, the spacing of the increments.

When using an analogue scale (e.g., a ruler), the error is half the smallest increment (e.g., 0.5 mm for a standard ruler). When using a digital scale, the error (unless otherwise stated on the instrument) should be taken as one unit of the lowest significant figure shown (e.g., if the device shows `1.67` the error would be 0.01). This assumes that the instrument was precisely zeroed by e.g., precisely placing the start of an object to be measured at zero or taring a digital scale.

To understand the error in a digital scale, suppose we were to measure the mass of a 1.674 kg block using a digital scale and it showed `1.67 kg`. Due to our scale's resolution and our lack of knowledge of the device's implementation of e.g. rounding, we would only know that the mass was between 1.66 kg and 1.68 kg — hence we assign an uncertainty of ± 0.01 kg.

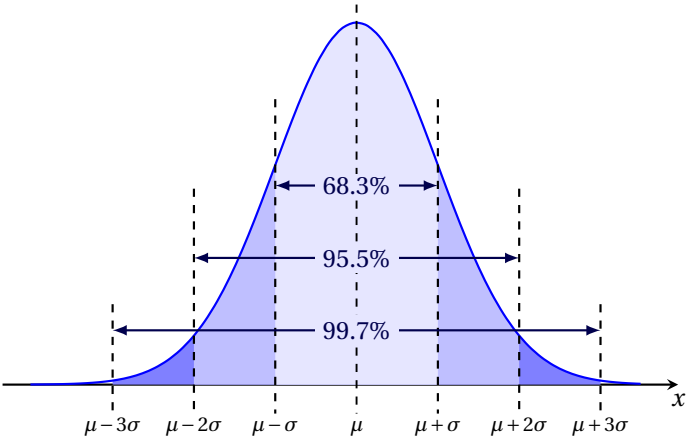


Figure 4: *The Gaussian or normal distribution with mean μ and standard deviation σ.*

Resolution errors are systematic errors. If we repeated a measurement read from a scale, we would make the same resolution error over and over again.

4.2 Mean and standard error

Suppose we make repeated measurements of a parameter in a case where we expect a statistical error. We expect these repeated measurements to show statistical variation. Indeed, these measurements can be modeled as independent draws from a data generating process that depends on the true un-

known value of the parameter. E.g., we could be measuring the decay rate of a radioactive sample by counting photons. The observed number of photons is a draw from a Poisson that depends on the true unknown value of the decay rate.

Unfortunately, we rarely know exactly what that data generating process is. We can, however, learn about it from the data itself. Suppose we take n repeated measurements, x_i, of a quantity of interest, x. We can estimate x from the mean of the repeated measurements and estimate the statistical error in x by the variation in the repeated measurements.

Indeed, we can estimate the true unknown value of x by,

$$\hat{x} = \bar{x} \pm \text{S.E.}$$

where the sample mean,

$$\bar{x} = \frac{1}{n} \sum_{i=1}^{n} x_i$$

and the standard error (S.E),

$$\text{S.E.} = \sqrt{s^2/n}$$

where the sample variance,

$$s^2 = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \bar{x})^2.$$

Notice that the standard error decreases as $1/\sqrt{n}$ for statistical errors. The error does not decrease at all with n for systematic errors.

Lastly, we can even estimate the error on the error from the data. We roughly expect that,

$$\Delta s/s \approx \frac{1}{\sqrt{2(n-1)}}$$

Thus, justifying more than one significant figures in a quoted standard error requires more than $n = 5,000$ measurements.

By the central limit theorem, we typically expect the sample mean to follow a Gaussian distribution — see fig. 4 — with a standard deviation estimated by the standard error.

4.3 Error propagation

Suppose we wish to propagate uncertainties in x, y, z, ... to the function $f(x, y, z, \cdots)$. We can approximate it by

$$(\Delta f)^2 \approx \left(\frac{\partial f}{\partial x}\right)^2 (\Delta x)^2 + \left(\frac{\partial f}{\partial y}\right)^2 (\Delta y)^2 + \left(\frac{\partial f}{\partial z}\right)^2 (\Delta z)^2 + \cdots$$

This neglects possible correlations between sources of error and relies on a first-order Taylor expansion of the function f.

Applying this formula, we find the following cases,

$f(x,y)$	Δf
ax	$\lvert a\rvert\Delta x$
$x+y$	$\sqrt{(\Delta x)^2+(\Delta y)^2}$
$x-y$	$\sqrt{(\Delta x)^2+(\Delta y)^2}$
xy	$\lvert xy\rvert\sqrt{(\Delta x/x)^2+(\Delta y/y)^2}$
x/y	$\lvert x/y\rvert\sqrt{(\Delta x/x)^2+(\Delta y/y)^2}$
x^a	$\lvert ax^{a-1}\rvert\Delta x$
e^{ax}	$\lvert a\rvert e^{ax}\Delta x$
$\ln x$	$\Delta x/x$, for $x>0$

The particular case $f(x,y)=x+y$, giving

$$\Delta f=\sqrt{(\Delta x)^2+(\Delta y)^2}$$

is known as adding errors in quadrature.

4.4 Tests

As well as measuring things, we often want to test principles or theories in experiments. Alas, because of experimental uncertainties, we cannot say definitively whether data rules out a theory.

Suppose we investigate a collision between blocks and measure a change in total momentum

$$\Delta p=(1.7\pm1.4)\,\mathrm{kg\,m\,s^{-1}}$$

Conservation of momentum predicts that $\Delta p=0$. Do we rule it out? Was our measurement consistent with zero? A simple

check is to count the number of standard errors between our measurement and the prediction,

$$Z = |1.7 - 0|/1.4 = 1.2$$

We refer to this as a deviation of 1.2σ. Looking at fig. 4, we see that deviations of 1.2σ or more occur at a rate of about 20% by chance alone. Thus our result does not appear inconsistent with conservation of momentum. Only deviations of more than 2σ are usually interpreted as evidence against a theory.

Tests and the interpretations of tests, however, are extremely contentious, and papers and books continue to be written about them. Regardless of methodology, however, as physicists before ruling out a theory, we must carefully consider systematic errors in our experiment that could cause a deviation.

5 Presentation

We want to communicate our scientific results clearly and concisely, so that they are not misinterpreted. Our results are shown in text, such as the final measurement of quantity or a test, and in tables and figures, such as the data.

5.1 Measurements

Measurements should be reported:

1. Usually in standard form, especially if particularly big or small
2. Always with an uncertainty
3. Always with an appropriate unit (usually an SI unit)
4. Always with appropriate significant figures
5. Always with consistent decimal places in the measurement and uncertainty

Usually, uncertainties are hard to quantify exactly and should be reported with 1 significant figure, unless you have justifica-

tion for more.

For example, consider the reported measurement of Planck's constant,

$$h = (6.9 \pm 0.1) \times 10^{-34}\,\mathrm{J\,s}$$

Does it match our recommendations?

1. In standard form? Yes.
2. With an uncertainty, Yes, ± 0.1.
3. With an appropriate unit? Yes, J s.
4. With appropriate significant figures? Yes, error was shown with 1 significant figure.
5. With consistent decimal places in the measurement and uncertainty? Yes, the error and measurement were both shown to one decimal place.

Note that we placed the measurement and uncertainty inside brackets. This makes it much easier to read than

$$h = 6.9 \times 10^{-34} \pm 0.1 \times 10^{-34}\,\mathrm{J\,s}$$

Measurements aren't always reported in standard form or using an SI unit. E.g., the mass of the top quark,

$$172 \pm 2\,\mathrm{GeV}$$

is reported using a common unit in fundamental physics, the electron-Volt, and since 172 isn't especially big or small, it isn't written in standard form as 1.72×10^2.

5.2 Tables

Often, we use tables to show important data that cannot be represented by a figure or that we need to show numerically. The rules are for data entries are the same as those in section 5.1. Additionally, for clear and efficient communication,

1. Column headings should be concise and clear, and must show units
2. Appropriate units for data
3. Table must have a number and a caption that concisely explains the contents — table should be understandable without referring to main text
4. Notation and abbreviations should be consistent with main text
5. Don't repeat data presented in text or in figures, and avoid repeating entries in the table
6. Arrange columns to best show your results
7. Use lines and borders sparingly; vertical lines are rarely necessary
8. Particularly large tables should appear in appendices or supplementary material

We show an example table in table 1.

5.3 Figures

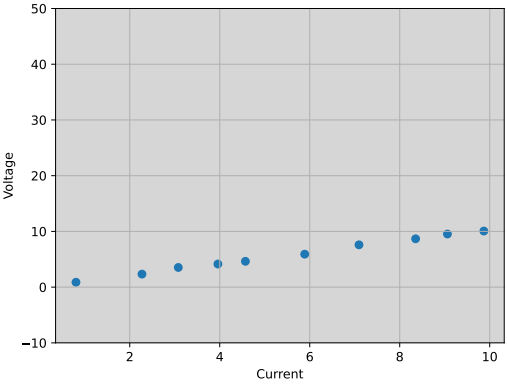
Figures are ways of visualizing results or trends in scientific data (e.g., graphs), scientific data itself (e.g., photos), an experimental setup (e.g., a photo or a schematic).

Data visualization is something of an art and entire books have been written about it. Think carefully about the purpose of a figure and the scientific message you wish to convey.

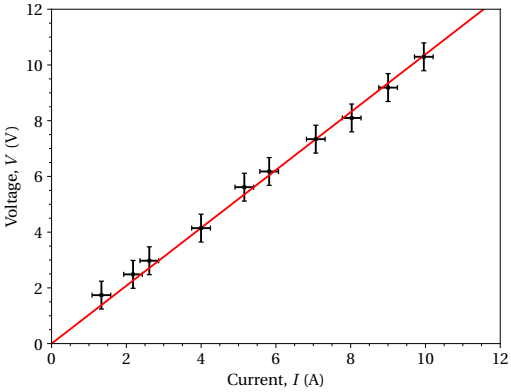
We focus on graphs here:

1. Axes labels should be concise and clear, and must show units
2. Appropriate units for data
3. Figure must have a number and a caption that concisely explains the contents — figure should be understandable without referring to main text
4. Notation and abbreviations should be consistent with main text
5. Experimental data must be shown with error bars
6. Appropriate tick locations and scale — if the data spans several orders of magnitude, consider a logarithmic scale
7. Distinguish different data sets by color and marker or line style
8. Use a readable font size (especially if showing figure in a presentation)
9. As a general rule of thumb, maximize "data ink" to ink — remove superfluous grids, backgrounds etc

We show an example figure in fig. 5.



(a) *This figure violates our recommendations.*



(b) *This figure follows our recommendations.*

Figure 5: *The same dummy dataset plotted in two different ways.*

M (kg)	L (cm)	T
0.252 ± 0.001	9.990 ± 0.01	5.9242
0.2524 ± 0.001	20.051 ± 0.01	9.5983
0.252 ± 0.001	29.970 ± 0.01	12.2232
0.5211 ± 0.001	9.99 ± 0.01	6.9113
0.521 ± 0.001	20.050 ± 0.01	9.11
0.52100 ± 0.001	29.969 ± 0.01	12.4
0.988 ± 0.001	9.99 ± 0.01	6.4
0.988 ± 0.001	20.05 ± 0.01	9.7
0.988 ± 0.001	29.9700 ± 0.01	11.91

(a) *This table violates our guidelines.*

Length (cm)	Mass (kg)	Period (s)
9.99 ± 0.01	0.252 ± 0.001	5.9 ± 0.5
	0.521 ± 0.001	6.9 ± 0.3
	0.988 ± 0.001	6.4 ± 0.4
20.05 ± 0.01	0.252 ± 0.001	9.6 ± 0.2
	0.521 ± 0.001	9.1 ± 0.3
	0.988 ± 0.001	9.7 ± 0.3
29.97 ± 0.01	0.252 ± 0.001	12.2 ± 0.3
	0.521 ± 0.001	12.4 ± 0.4
	0.988 ± 0.001	11.9 ± 0.3

(b) *This table follows our guidelines.*

Table 1: *The same dummy dataset tabluated in two different ways.*